

Ahmadreza Attarpour

+1 (437) 559-7779 | a.attarpour@mail.utoronto.ca | <https://www.linkedin.com/in/ahmadreza-attarpour>

AI-DRIVEN PIPELINES | MEDICAL IMAGE and SIGNAL ANALYSIS | HEALTHCARE

AI Scientist with over 9 years of experience at the intersection of AI/ML, neuroscience, and healthcare. Expert in developing and deploying end-to-end AI pipelines for large-scale biomedical and microscopy data analysis, including foundation deep learning models and agentic AI systems leveraging vision-language representations. Experienced in designing ML pipelines robust to distribution shift, noisy data, and model uncertainty in decision settings. Demonstrated impact includes 2 patent filings and 6 peer-reviewed publications (including first-author Nature Methods), as well as 10+ international conference presentations. Committed to translating advanced ML methodologies into clinically and scientifically impactful solutions through interdisciplinary leadership.

RELEVANT EXPERIENCE

Highlighted Projects

Foundation Deep Learning Model for CryoET | *Python, PyTorch* Dec. 2025 – Present

- Working on a foundation DL model for object detection and segmentation in CryoET 3D tomograms.
- Gathering around 1600 3D tomogram and generated more than 350,000 3D patches for pretraining using 3DINO self-supervised learning approach.
- Finetune the model on different downstream tasks and compare against the baseline models (In progress).

Agentic AI System for Digital Pathology | *Python, PyTorch, LangChain, VLM* Nov. 2025 – Present

- Engineered a multi-modal agentic framework orchestrating custom tools for automated WSI analysis, including Visual Question Answering (VQA), report generation, and zero-shot classification and ROI segmentation.
- Integrated state-of-the-art pathology foundation models (SlideChat, TITAN, CONCH, MUSK) to enable context-aware reasoning over gigapixel-scale images.
- Implemented a human-in-the-loop interface with a hierarchical k-means algorithm to iteratively refine model predictions for complex classification tasks, ensuring interpretability and clinical alignment.

MAPL3 (Mapping Axonal Projection in Light Sheet Microscopy in 3D) | *Python, Bash* Jan. 2023 – May. 2025

- Developed a 3D computational pipeline for identifying **individual** axonal fibers and their circuitry brain-wide.
- Pioneered a novel DL architecture that combines **convolutional neural networks and vision transformers**.
- Deployed **self-supervised** approaches to improve model generalizability and reduce dependency on labeled data.
- Designed and implemented **parallelized image processing** ([here](#)) pipelines for pre- and post-processing.

ACE (AI-based Cartography of Ensembles) | ([here](#)) *Python, Fiji, Bash, Docker, GIT* Jan. 2020 – Jan. 2025

- Designed and implemented an **end-to-end** 3D pipeline for mapping **local** cell activity in tera-voxel scale (trillions of voxels; **~1TB per sample**) light sheet microscopy datasets.
- Engineered an ensemble of **convolutional and vision transformer-based** for soma segmentation.
- Integrated the Monte Carlo dropout technique to provide probabilistic **model uncertainty** estimation.
- Optimized cluster-wise permutation statistical analysis for high-dimensional whole-brain microscopy data.

Foundation Deep Learning Model for Light Microscopy | *Python, Bash, GIT* Jan. 2023 – Dec. 2025

- Working on a foundation DL model for object segmentation in microscopy datasets.
- Gathering around **3 million** 3D image patches from different centers and modalities for pretraining the model using 3DINO approach.
- Finetuning the model for axon, soma, vascular segmentation in and compare against state-of-the-art models.

Pancreas Multi-Task Learning with nnUNetV2 | ([here](#)) *Python, Pytorch, GIT* Apr. 2025 – Jun. 2025

- Developed a DL pipeline for **joint pancreas segmentation and subtype classification** on abdominal CT using a modified nnUNetV2 backbone.
- Integrated a custom classification head with multi-scale feature adapters and fusion modules to process encoder outputs alongside segmentation decoding.
- Achieved average Dice of **0.90 (pancreas)** and **0.60 (lesion)** and **macro F1 of 0.81** on test; reduced inference time by **24%** via optimized prediction loop.

Development of a System for Recording and Analyzing Pulse Signals | *MATLAB* Sept. 2015 – June 2016

- Designed a system for **continuous** blood pressure measurement using PPG recorded from the wrist and fingertip.
- Developed a **Multi-Layer Perceptron (MLP)** neural network to estimate systolic and diastolic blood pressure (SBP/DBP) with mean absolute errors of 4.94 mmHg and 4.03 mmHg, respectively.
- Implemented feature selection algorithms to rank and refine the most significant features, including **moving-backward and genetic optimization**, achieving optimal performance with reduced feature sets.

Work Experience

AI Scientist

Nov. 2025 – Present

University Health Network (UHN) & Peter Munk Cardiac Centre (PMCC)

Toronto, Canada

- Leading the development of **agentic AI workflows** and **foundation models** for large-scale biomedical data.
- Collaborating with interdisciplinary teams of clinicians and researchers to translate novel ML research into deployable solutions for biomedical data.
- Serve in advisory and mentorship roles across multiple academic and applied AI projects (UHN AIHub, University of Toronto), including supervising interns and PhD students, planning experiments, debugging ML/DL pipelines, and paper writing.

Software Engineer and Programmer

Feb. 2017 – Jul. 2019

RayanMindware Co.

Isfahan, Iran

- Developed algorithms for time-frequency EEG signal analysis using MATLAB.
- Optimized MATLAB scripts and chipset code in C++ to enhance performance.
- Developed and tested wireless data transfer systems for EOG signals.

Other Activities

Research Supervision

2021 – Present

University of Toronto & University Health Network

Toronto, CA

- Supervised four undergraduate thesis/summer students on their research projects.
- Provided mentorship on ML/DL techniques, data analysis, and scientific writing.

Reviewer

Oct. 2018 – Present

Various Journals and Councils

- Reviewed articles for NeuroImage and Biomedical Signal Processing and Control Journals (Oct. 2018 – present).
- Reviewed grant proposals on non-invasive deep brain stimulation for the Cognitive Sciences and Technologies Council, Tehran, Iran (Nov. 2019).

PATENTS AND DISCLOSURES

System and Method for Unbiased Mapping of Local Neuronal Activity and Connectivity 2024

A. Attarpour, B. Stefanovic, M. Goubran, Sunnybrook research institute
Under Review, connected to a US provisional application

Canada

A System for Cuff-less Continuous Measurement of Blood Pressure from Pulse Signals 2018

A. Attarpour, A. Mahnam, University of Isfahan

Iran

Patent Number: 95585, International Classification: A61B; G01N; A61F; A61K

SELECTIVE PUBLICATIONS (GOOGLE SCHOLAR LINK)

A. Attarpour, M. Raffee, T. Xu, J. Osmann, S. Patel, F. Yu, B. Au, A. Clappison, M. Biparva, R. Zhu, A. Crow, B. Gharagoz, M. Rozak, I. Aubert, J. McLaurin, K. Deisseroth, “Quantitative profiling of whole-brain connectomes at single-axon resolution using deep learning and high-resolution light sheet microscopy,” ***Under review in Nature Methods***, Dec. 2025

A. Attarpour, J. Osmann, A. Rinalid, T. Qi, N. Lal, S. Patel, M. Rozak, F. Yu, N. Cho, J. Squari, J. McLaurin, M. Raffee, K. Deisseroth, L. Ye, B. Stefanovic, M. Goubran, “A deep learning pipeline for 3D brain-wide mapping of local neuronal ensembles in tera-voxel light sheet microscopy,” ***Nature Methods***, Jan. 2025, <https://doi.org/10.1038/s41592-024-02583-1>

A. Attarpour, J. Ward, J. Chen, “Vascular Origins of Low-Frequency Oscillations in the Cerebrospinal Fluid Signal in Resting-State fMRI: Interpretation Using Photoplethysmography,” ***Human Brain Mapping***, 2021, <https://doi.org/10.1002/hbm.25392>.

F. Karimi*, A. Attarpour*, R. Amirfattahi, A. Zeidabadinezhad, “Computational Analysis of Non-Invasive Deep Brain Stimulation Based on Interfering Electric Fields,” ***Physics in Medicine & Biology***, 2019, <https://doi.org/10.1088/1361-6560/ab5229>. **Equal contribution as first authors.*

A. Attarpour, A. Mahnam, A. Aminitabar, H. Samani, “Cuff-less continuous measurement of blood pressure using wrist and fingertip photoplethysmograms: Evaluation and feature analysis,” ***Biomedical Signal Processing and Control***, 2019, <https://doi.org/10.1016/j.bspc.2018.12.006>.

SELECTIVE PRESENTATIONS

IGT x ImNO Joint Symposium, Toronto, CA <i>A deep learning pipeline for 3D brain-wide mapping of local neuronal ensembles</i>	Mar. 2025
Keynote speaker at UBC Tissue Clearing and Expansion workshop, Vancouver, CA <i>ACE – AI based Cartography of Ensembles</i>	May. 2024
Poster at SFN conference, Chicago, US <i>An end-to-end deep learning pipeline for 3D mapping of laminar neuronal activity in light sheet data</i>	Oct. 2024
VAST Conference, Montreal, CA <i>A deep learning network for 3D segmentation of neurons in tera-voxel light sheet fluorescence microscopy</i>	May. 2023
Nanosymposium presentation session at SFN conference, Washington, US <i>A deep learning pipeline for 3D mapping of neuronal activity in tera-voxel light sheet microscopy data</i>	Nov. 2023
International Society for Magnetic Resonance in Medicine (ISMRM), Online <i>Vascular origins of low-freq. oscillations in the CSF in resting-state fMRI: Interpretation using PPG</i>	May. 2021

TECHNICAL SKILLS

Programming Languages: Python, C/C++, MATLAB, Bash, Java
AI/ML Frameworks: PyTorch, Keras, Scikit-learn
Medical Imaging and Signal Analysis: FSL, ITKsnap, Freesurfer, Fiji (ImageJ), Nibabel, Brainrender
Libraries: NumPy, SciPy, Pandas, Matplotlib
Developer Tools: Git, Docker, Jupyter, Spyder, Visual Studio, VS Code
Research Techniques: Machine learning (deep learning), Agentic system, Statistical analysis, complex large microscopy, MR, CT, and Ultrasound data analysis, Electrophysiology

EDUCATION

University of Toronto <i>Ph. D. in Medical Biophysics, Department of Medical Biophysics, Faculty of Medicine</i>	Ontario, CA Sep. 2020 – Nov. 2025
Isfahan University of Technology <i>M.Sc. in Electrical Engineering, Department of Electrical and Computer Engineering</i>	Isfahan, Iran Sept. 2016 — Jul. 2019
University of Isfahan <i>B.Sc. in Biomedical Engineering (Bioelectronics), Department of Biomedical Engineering</i>	Isfahan, Iran Sept. 2012 — Sept. 2016

PROFESSIONAL DEVELOPMENT

Natural Language Processing Short Course <i>AI4PH</i>	Jan. 2025 – March. 2025 Online
Python for High-Performance Computing Workshop <i>SHARCNET Training</i>	Oct. 2024 Online
Author, Review Paper <i>Advances in Recombinant Adeno-Associated Virus Vectors for Neurodegenerative Diseases</i>	Jan. 2023 – Apr. 2023 Biomedicines
CIHR Institute of Aging Summer Program in Aging (SPA) <i>Awarded by the Brenda Strafford Center on Aging to attend this program.</i>	May. 2022 – Jul. 2022 University of Calgary, Alberta, CA
Deep Learning Foundation Course <i>SHARCNET Summer School.</i>	Aug. 2021 Online
Machine Learning for the Cognitive Neurosciences Workshop <i>McGill Center for Integrative Neuroscience, HBHL's NeuroHub, ReproNim, and UNIQUE</i>	Jan. 2021 Online
Writing Workshops <i>Graduate Center for Academic Communication, University of Toronto</i>	May. 2021 – June. 2021 Online

AWARDS & HONORS

Medical Biophysics Excellence Award <i>Sunnybrook Research Institute, Physical Sciences Platform</i>	Sept. 2020 – Aug. 2021 & Jan. 2022 – Aug. 2022 Ontario, CA
VAST Doctoral Scholarship Award <i>VAST Health Research Training Program - University of Calgary</i>	Sept. 2022 — Aug. 2023 Alberta, CA
Institute for Cognitive Science Studies Fellowship <i>Outstanding research in the field of Deep Brain Stimulation</i>	Sept. 2017 — Jul. 2019. Tehran, Iran